| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 10(d) | Ruled line from (9 40, 0) to (10 20, 14) | **2** | **M1** for $\frac{\textit{their}\,\textbf{(a)}}{21}$ oe or a parallel line |
| 11 | 27 | **1** | |
| 12 | −5 cao final answer | **2** | **M1** for $5x - 3x = -2 - 8$ oe or better |
| 13(a) | 20 | **1** | |
| 13(b) | –11 | **1** | |
| 14 | 6 final answer | **1** | |
| 15 | Correct net | **3** | **B2** for 3 correct extra faces in correct places or **B1** for 1 correct extra face in correct place |
| 16(a) | $5a - b$ final answer | **2** | **B1** for $5a$ or $-b$ in final answer or for $5a - b$ seen then spoilt |
| 16(b)(i) | $3(2x + 5y)$ final answer | **1** | |
| 16(b)(ii) | $xy(x - 5)$ final answer | **2** | **B1** for $x(xy - 5y)$ or $y(x^2 - 5x)$ or $xy(x - 5)$ seen then spoilt |
| 17 | $\frac{40+20}{2\times6}$ | **M1** | |
| | 5 nfww | **A1** | If 0 scored **SC1** for 3 correct roundings or for all correct but with trailing zeros |
| 18(a) | 50 alternate angles | **2** | **B1** for each |
| 18(b) | 110 corresponding angles | **2** | **B1** for each |
| 18(c) | 60 | **2** | **M1** for 180 – (180 – 110) – 50 oe or 70 marked as third angle in triangle |
| 19(a) | Correct tree diagram | **3** | **B1** for $\frac{1}{5}$ oe **and** $\frac{4}{5}$ seen oe<br>**M1** for either $\frac{1}{5}$ **or** $\frac{4}{5}$ in the correct place in the **first** branch<br>If 0 scored, **SC1** for their two probabilities adding to 1 in the **first** branch |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 19(b) | $\frac{16}{25}$ oe | **2** | **FT** *their* **(a)** probabilities provided 0 < each of *their* $\frac{4}{5}$ < 1<br>**M1** for *their* $\frac{4}{5}$ × *their* $\frac{4}{5}$ |
| 20(a) | 2 points accurately plotted | **1** | |
| 20(b) | Negative | **1** | |
| 20(c)(i) | Correct ruled line of best fit | **1** | |
| 20(c)(ii) | 275 | **1** | **FT** *their* straight line of best fit with negative gradient |
| 21 | 51 | **2** | **M1** for 90 – 39 oe<br>or angle *ABC* identified as 90 |
| 22(a) | 3 | **1** | |
| 22(b) | 360 | **2** | **B1** for answer 360*k*, *k* > 1<br>or **M1** for $2^3 \times 3^2 \times 5$ oe<br>or for 24, 48, 72… **and** 45, 90, 135… seen |
| 23(a) | Correct rotation, points at (1, –1), (3, –1) and (1, –4) | **2** | **B1** for correct rotation 90° anti-clockwise or correct rotation with incorrect centre |
| 23(b) | Translation<br>$\begin{pmatrix} -6 \\ -4 \end{pmatrix}$ | **2** | **B1** for each |
| 23(c) | Enlargement<br>[scale factor] 2<br>[centre] (0, -2) | **3** | **B1** for each<br>Angle mentioned |
| 24 | $3\frac{1}{12}$ nfww | **3** | **B1** for $\frac{4}{3}$ **or** $\frac{7}{4}$ oe<br>or for $2 + \frac{1}{3} + \frac{3}{4}$ oe<br>**M1** for $\frac{16}{12}$ **and** $\frac{21}{12}$ or $\frac{4}{12}$ **and** $\frac{9}{12}$ |
| 25(a) | $3.25 \times 10^4$ | **1** | |
| 25(b) | [0].0056 | **1** | |